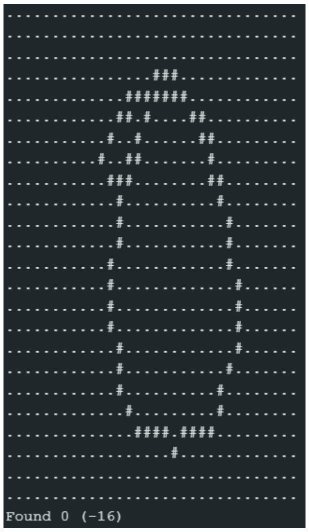
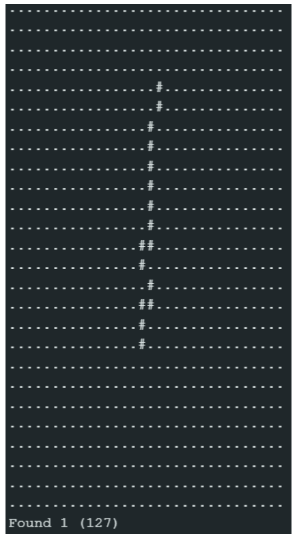
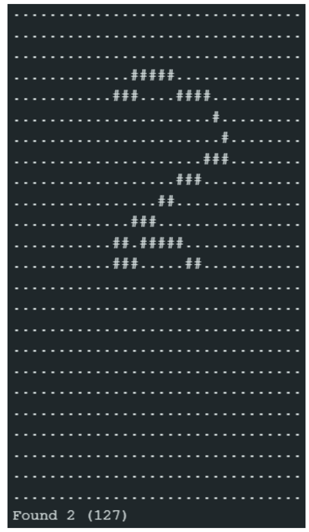
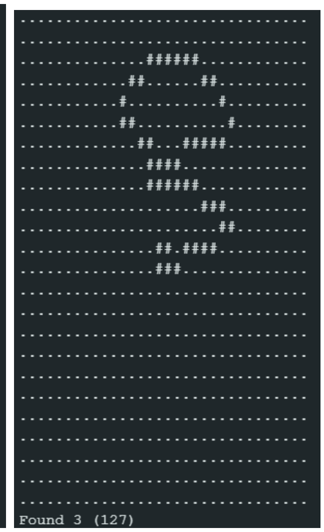
Lab 7

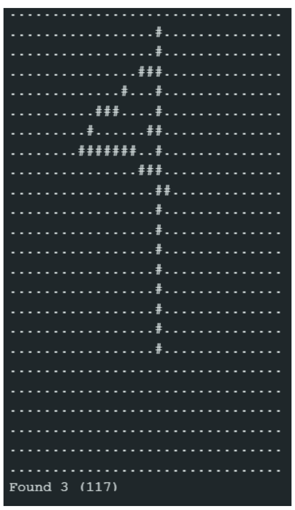
Task 1:

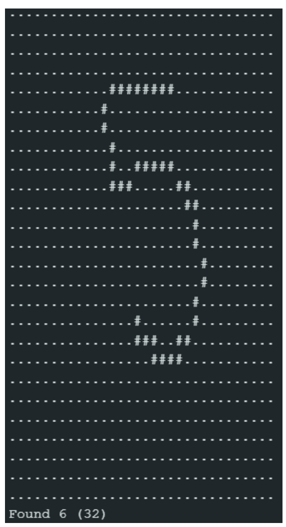

Found 0 (-16)

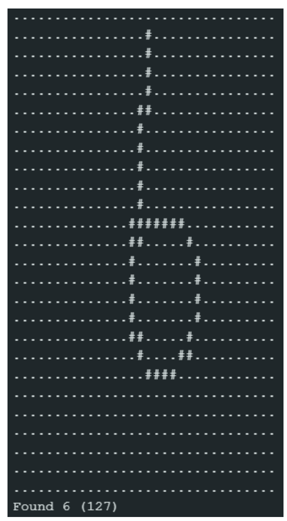

Found 1 (127)

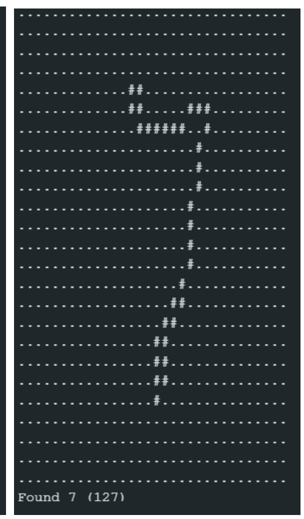

Found 2 (127)

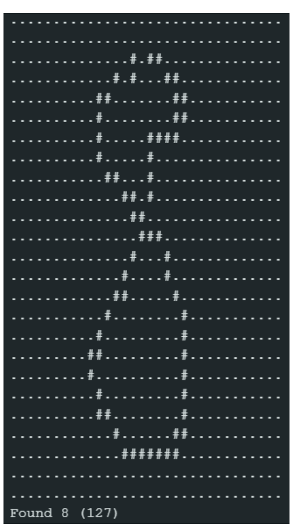

Found 3 (127)

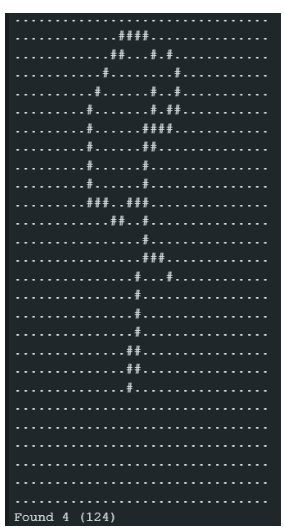

Found 3 (117)


Found 6 (32)

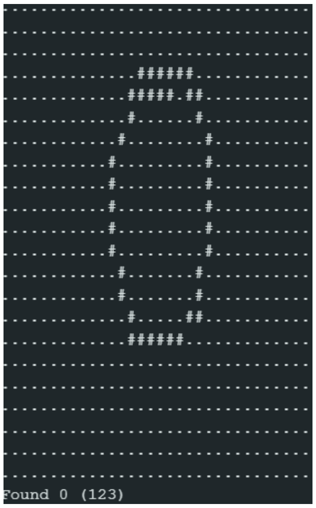

Found 6 (127)

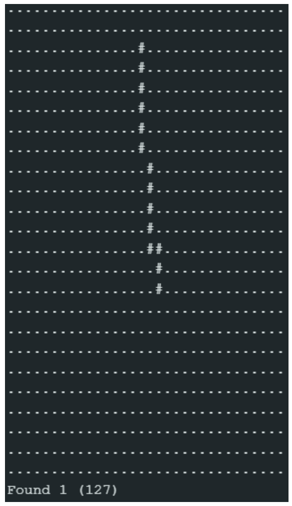

Found 7 (127)

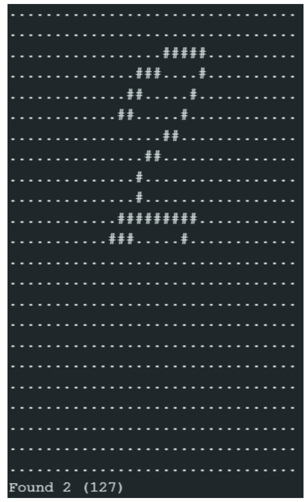

Found 8 (127)

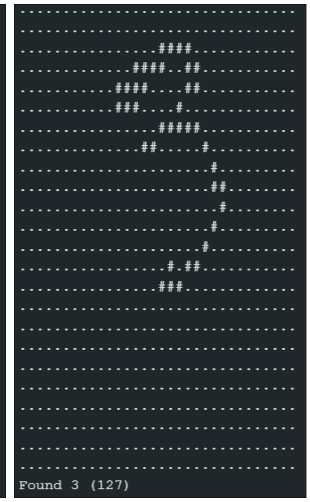

Found 4 (124)

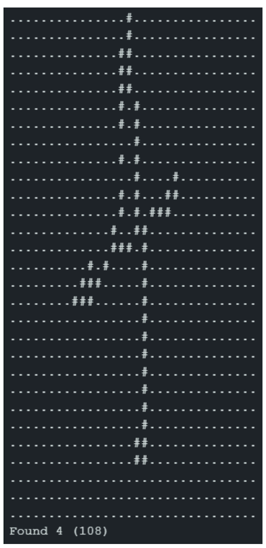
Task 3:


Found 0 (123)

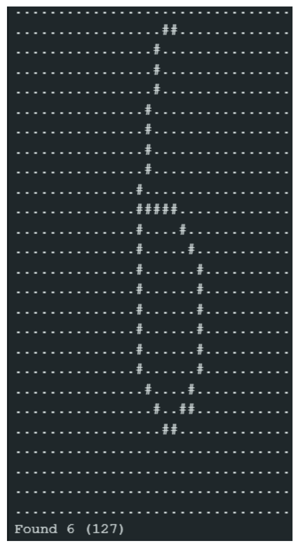

Found 1 (127)

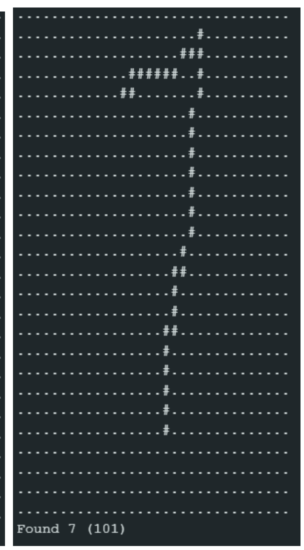

Found 2 (127)

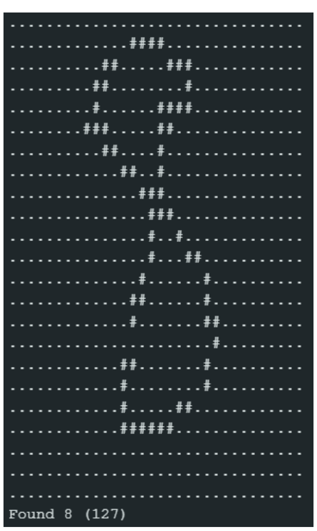

Found 3 (127)

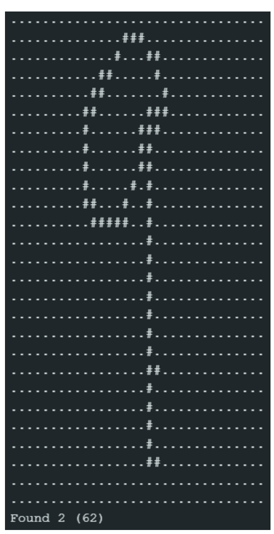

Found 4 (108)


Found 0 (107)


Found 6 (127)


Found 7 (101)


Found 8 (127)


Found 2 (62)

Task 4:

The fine-tuned model in Task 3 performed better than the baseline model in Task 1. This improvement was shown by the higher classification metrics, including F1 score, recall, and precision, across the digit classes. The difference was also noticeable when the models were deployed on the Arduino: the fine-tuned Task 3 model produced more accurate and consistent predictions, correctly identifying 8 out of 10 digits, while the baseline model only identified 7 out of 10. Overall, these results suggest that fine-tuning improved the model's ability to distinguish between gestures, especially when the gestures were distinct. However, the model still struggled more when gestures were too similar, indicating that fine-tuning helped but did not completely eliminate classification confusion.